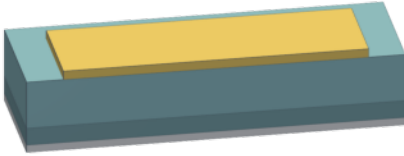


4H-SiC power-device structures

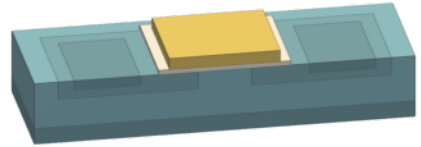
DSL-authored solids; doped shades distinguish SiC regions

(a) Vertical Schottky diode



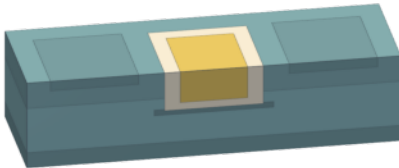
front metal / n-drift / n+ substrate / backside metal
Cooper & Agarwal, Proc. IEEE 90 (2002)

(b) Planar double-implanted MOSFET



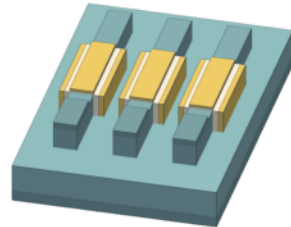
replacement p-body + n+ source + surface gate
Vathulya & White, IEEE TED 47 (2000)

(c) Shielded trench MOSFET



nested oxide/gate fills above a p+ shield implant
Zhou et al., IEEE TED 64 (2017)

(d) SiC tri-gate concept



three fins with non-overlapping oxide-isolated wrap gates
Ramamurthy et al., IEEE EDL 42 (2021)